SQL Query Implementation for E-Commerce Database

Objectives :

- To perform SELECT, WHERE, ORDER BY, and DISTINCT queries.
- To search products based on price, category, and availability.
- To retrieve customer and product information.
- To apply filtering conditions using AND, OR, IN, and BETWEEN operators.
- To generate basic business reports using SQL queries.

1. Basic SQL Queries :

1.1 SELECT Query :

Display all Products.

```
SQL> SELECT * FROM Product;

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
       101 Laptop                                                      1
         10      55000         20
Dell Laptop

       102 Smartphone                                                  2
         20      25000         30
Android Smartphone

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------

       103 Headphones                                                  3
         30       2500         50
Wireless Headphones

       104 Keyboard                                                    4
         30       1500         40
```

```
PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
USB Keyboard

       105 Mouse                                                       5
         30        800         60
Wireless Mouse

       106 Monitor                                                     6

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
         10      12000         15
LED Monitor


6 rows selected.
```

Display selected product details.

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product;

PRODUCT_ID PRODUCT_NAME                                          PRICE
---------- -------------------------------------------------- ----------
       101 Laptop                                              55000
       102 Smartphone                                          25000
       103 Headphones                                           2500
       104 Keyboard                                             1500
       105 Mouse                                                 800
       106 Monitor                                             12000

6 rows selected.
```

1.2 WHERE Clause :

The WHERE clause is used to filter records based on a condition.

a) Products with price less than 1000

```
SQL> SELECT * FROM Product
  2  WHERE Price < 1000;

PRODUCT_ID PRODUCT_NAME                                    SELLER_ID
---------- ---------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
       105 Mouse                                                   5
        30        800         60
Wireless Mouse
```

b) Find products with price greater than 100

```
SQL> SELECT * FROM Product
  2  WHERE Price > 100;

PRODUCT_ID PRODUCT_NAME                                    SELLER_ID
---------- ---------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
       101 Laptop                                                  1
        10      55000         20
Dell Laptop

       102 Smartphone                                              2
        20      25000         30
Android Smartphone

PRODUCT_ID PRODUCT_NAME                                    SELLER_ID
---------- ---------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------

       103 Headphones                                              3
        30       2500         50
Wireless Headphones

       104 Keyboard                                                4
        30       1500         40
```

```
PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
USB Keyboard

       105 Mouse                                                       5
         30        800         60
Wireless Mouse

       106 Monitor                                                     6

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
         10      12000         15
LED Monitor


6 rows selected.
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

a) Lowest to highest price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID PRODUCT_NAME                                          PRICE
---------- -------------------------------------------------- ----------
       105 Mouse                                                 800
       104 Keyboard                                             1500
       103 Headphones                                           2500
       106 Monitor                                             12000
       102 Smartphone                                          25000
       101 Laptop                                              55000

6 rows selected.
```

b) Highest to lowest price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID PRODUCT_NAME                                         PRICE
---------- -------------------------------------------------- ----------
       101 Laptop                                              55000
       102 Smartphone                                          25000
       106 Monitor                                             12000
       103 Headphones                                           2500
       104 Keyboard                                             1500
       105 Mouse                                                 800

6 rows selected.
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories

```
SQL> SELECT DISTINCT Category_ID
  2  FROM Product;

CATEGORY_ID
-----------
         10
         20
         30
```

2. Search Products :

2.1 Search Products Based on Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 100 AND 1000;

PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- -------------------------------------------------- ----------
       105 Mouse                                               800
```

This displays products whose price is between ₹100 and ₹1000

2.2 Search Products Based on Category :

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE ID = 1;

        ID NAME                 AGE ADDRESS
---------- -------------------- ---------- ------------------------------
    SALARY
----------
         1 Ramesh                32 Ahmedabad
     30000
```

3. Retrieve Customer Information :

Display all customers

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE ID = 1;

        ID NAME                 AGE ADDRESS
---------- -------------------- ---------- ------------------------------
    SALARY
----------
         1 Ramesh                32 Ahmedabad
     30000


SQL>
SQL> SELECT *
  2  FROM Customer;

        ID NAME                 AGE ADDRESS
---------- -------------------- ---------- ------------------------------
    SALARY
----------
         1 Ramesh                32 Ahmedabad
     30000

         2 Khilan                25 Delhi
     30000

         3 Kaushik               23 Kota
     30000
```

```
        ID NAME                 AGE ADDRESS
---------- -------------------- ---------- ------------------------------
    SALARY
----------
         4 Chaitali              25 Mumbai
     75000

         5 Hardik                27 Bhopal
     80000

         6 Komal                 22 MP
     55000


6 rows selected.
```

Search for a particular customer

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE ID = 1;

        ID NAME                           AGE ADDRESS
---------- -------------------- ---------- ------------------------------
    SALARY
----------
         1 Ramesh                         32 Ahmedabad
     30000
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE Price < 1000
  4  AND Category_ID = 30;

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
       105 Mouse                                                         5
        30        800         60
Wireless Mouse
```

4.2 OR Condition

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID = 20
  4     OR Category_ID = 30;

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
       102 Smartphone                                                  2
         20      25000         30
Android Smartphone

       103 Headphones                                                  3
         30       2500         50
Wireless Headphones

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------

       104 Keyboard                                                    4
         30       1500         40
USB Keyboard

       105 Mouse                                                       5
         30        800         60

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
Wireless Mouse
```

4.3 IN Operator

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE Category_ID IN (10,20,30);

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
       101 Laptop                                                      1
         10      55000         20
Dell Laptop

       102 Smartphone                                                  2
         20      25000         30
Android Smartphone

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------

       103 Headphones                                                  3
         30       2500         50
Wireless Headphones

       104 Keyboard                                                    4
         30       1500         40
```

```
PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
USB Keyboard

       105 Mouse                                                       5
         30        800         60
Wireless Mouse

       106 Monitor                                                     6

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
         10      12000         15
LED Monitor


6 rows selected.
```

4.4 BETWEEN Operator

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE Price BETWEEN 100 AND 1000;

PRODUCT_ID PRODUCT_NAME                                        SELLER_ID
---------- -------------------------------------------------- ----------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
       105 Mouse                                                       5
        30        800         60
Wireless Mouse
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID PRODUCT_NAME                                          PRICE
---------- -------------------------------------------------- ----------
       101 Laptop                                              55000
       102 Smartphone                                          25000
       106 Monitor                                             12000
       103 Headphones                                           2500
       104 Keyboard                                             1500
       105 Mouse                                                 800

6 rows selected.
```

5.2 Customer Order Report

```
SQL> SELECT C.ID, C.NAME, O.Order_ID,
  2         O.Order_Date, O.Total_Amount
  3  FROM Customer C
  4  JOIN Orders O
  5  ON C.ID = O.Customer_ID;

        ID NAME                 ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
         1 Ramesh                     1001 01-SEP-26         2098
         2 Khilan                     1002 03-SEP-26         1499
         3 Kaushik                    1003 05-SEP-26         2598
         4 Chaitali                   1004 07-SEP-26         1999
         5 Hardik                     1005 10-SEP-26          450
         1 Ramesh                     1006 12-SEP-26         1299
         3 Kaushik                    1007 15-SEP-26         1999
         5 Hardik                     1008 18-SEP-26         1049

8 rows selected.
```

5.3 Product Order Report

```
SQL> SELECT P.Product_ID, P.Product_Name,
  2         OD.Order_ID, OD.Quantity,
  3         OD.Unit_Price, OD.Subtotal
  4  FROM Product P
  5  JOIN Order_Details OD
  6  ON P.Product_ID = OD.Product_ID;

PRODUCT_ID PRODUCT_NAME                                       ORDER_ID
---------- -------------------------------------------------- ----------
  QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ----------
       101 Laptop                                                  201
         3        130        390

       102 Smartphone                                              201
         1        180        180

       103 Headphones                                              202
         2        350        700


PRODUCT_ID PRODUCT_NAME                                       ORDER_ID
---------- -------------------------------------------------- ----------
  QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ----------
       104 Keyboard                                                202
         1        450        450

       105 Mouse                                                   203
         2        250        500

       106 Monitor                                                 203
         1        120        120


6 rows selected.
```

5.4 Customer-wise Total Orders

```
SQL> SELECT C.ID, C.NAME,
  2         COUNT(O.Order_ID) AS Total_Orders
  3  FROM Customer C
  4  JOIN Orders O
  5  ON C.ID = O.Customer_ID
  6  GROUP BY C.ID, C.NAME;

        ID NAME                 TOTAL_ORDERS
---------- -------------------- ------------
         1 Ramesh                          2
         2 Khilan                          1
         3 Kaushik                         2
         4 Chaitali                        1
         5 Hardik                          2
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT C.ID, C.NAME,
  2         SUM(O.Total_Amount) AS Total_Purchase
  3  FROM Customer C
  4  JOIN Orders O
  5  ON C.ID = O.Customer_ID
  6  GROUP BY C.ID, C.NAME;

        ID NAME                 TOTAL_PURCHASE
---------- -------------------- --------------
         1 Ramesh                         3397
         2 Khilan                         1499
         3 Kaushik                        4597
         4 Chaitali                       1999
         5 Hardik                         1499
```

Conclusion :

Thus, SQL queries were successfully implemented for the E-Commerce Database. SELECT, WHERE, ORDER BY, DISTINCT, filtering conditions, and JOIN operations were used to retrieve and analyze product, customer, and order information. Basic business reports were generated to support efficient data management and decision-making.